

Esteban Sitek

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with **12+** years across healthcare, e-commerce, and fintech, building production data infrastructure in Snowflake, dbt, and modern data stacks. Proven expertise in medallion architectures, semantic modeling, data governance, and AI-ready platforms. Experienced in designing reliable pipelines, optimizing warehouse performance, and mentoring engineering teams to deliver scalable data products.

SKILLS

- Languages:** SQL, Python, JavaScript, TypeScript, Java, HTML, CSS
- Data & ML:** Snowflake, dbt, Coalesce, Fivetran, Apache Airflow, Spark, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, MLflow
- Backend:** Node.js, Express, FastAPI, Django, Spring Boot, PostgreSQL, Kafka, RabbitMQ
- Frontend:** React, Next.js, Vue.js, Tailwind CSS, Redux
- Database:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB
- Testing:** Jest, Pytest, Playwright, Cypress, Selenium
- Cloud & DevOps:** AWS, Google Cloud Platform, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, EC2, S3, RDS, CloudWatch
- Others:** Tableau, RESTful API, GraphQL, Git, GitHub, Jira, Confluence, Datadog, Prometheus, Grafana, LangChain, OpenAI API, Pinecone, ChromaDB

EXPERIENCE

Publicis Sapient Jun 2024–Present
Senior Software Engineer

- Led the design and implementation of enterprise data pipeline architecture supporting AI-driven decision-making platforms, establishing medallion patterns with **Snowflake**, **dbt**, and **Fivetran** across Bronze, Silver, and Gold layers that ingested 500M+ daily records.
- Architected semantic data models in **Snowflake** representing key business metrics and KPIs for financial services, healthcare, and retail verticals, enabling self-service analytics through **Tableau** while reducing ad hoc data requests by **42%**.
- Owned end-to-end data observability implementation, building freshness monitors, anomaly detection, and SLA tracking using **Python**, **Airflow**, and **Prometheus** that caught 18 production data quality incidents before they impacted downstream systems.
- Tuned **Snowflake** warehouse performance across 12+ concurrent business domain pipelines by optimizing clustering strategies, query patterns, and virtual warehouse sizing, reducing query latency from 8.2s to **1.9s** and monthly compute costs by **\$28K**.

- Designed and implemented AI-ready data products supporting LLM and RAG use cases, including vector embeddings ingestion pipeline with **Pinecone**, contextual retrieval patterns, and metadata enrichment for agent-based applications across three business units.
- Established data governance standards for PII classification, access control patterns, lineage documentation, and data contracts across 6 producer systems feeding the semantic layer, preventing unauthorized data exposure incidents.
- Mentored two senior analytics engineers through production transformation design reviews, code refactoring on complex **dbt** models, and hands-on incident resolution, elevating team query optimization capability by **35%**.
- Evaluated and led implementation of **Coalesce** transformation engine alongside existing **dbt** workflows, establishing testing frameworks, version control patterns, and deployment automation that improved deployment cycle time by **55%**.
- Partnered with Data Scientists and AI/ML teams to define data contracts at service boundaries, ensuring consistent schema definitions and preventing integration friction across analytics, product, and ML training pipelines.
- Documented platform capabilities, decision records, and data flow diagrams for scaling compliance and regulatory requirements across healthcare and financial services data domains.
- Built inbound and outbound data connectors using **Fivetran**, **Kafka**, and custom **Python** ETL jobs to reliably move data across 15+ external SaaS platforms, maintaining **99.7%** pipeline uptime.
- Implemented **Tableau** consumption layer optimization, working with BI team to define effective data source configurations and materialized views that improved dashboard load times from 12s to **2.1s** for executive reporting.

CareSimple

Oct 2022–May 2024

Senior Full Stack Engineer

- Spearheaded the healthcare data platform initiative, designing and building **Snowflake**-based medallion architecture across remote patient monitoring, chronic care management, and hospital-at-home data domains with **Fivetran** ingestion and **dbt** transformations.
- Implemented HIPAA-compliant data governance layer in **Snowflake**, defining PII classification, role-based access controls, and field-level encryption patterns that ensured regulatory compliance across 40+ clinical and operational datasets.
- Delivered semantic models representing patient cohorts, readmission risk scores, and population health metrics in **Snowflake**, exposing them through **Tableau** dashboards that enabled clinicians to identify high-risk patient populations in near real-time.
- Built data observability practice from scratch using **Python**, **Airflow**, and **Grafana**, establishing freshness SLAs, anomaly detection for clinical data anomalies, and incident response protocols that reduced undetected data quality issues by **89%**.
- Optimized **Snowflake** warehouse costs for healthcare analytics workloads by implementing dynamic scaling policies, result caching, and query tuning across 8 concurrent virtual warehouses, reducing monthly spend by **\$16K** while maintaining SLA compliance.

- Established data integration patterns for inbound medical device telemetry and EHR system feeds using **Kafka** and **Fivetran**, ensuring reliable delivery of real-time patient vitals and care events across clinical applications.
- Collaborated with AI/ML team to prepare training datasets for predictive readmission models, curating feature stores in **Snowflake** with proper documentation and validation logic that improved model reproducibility and reduced data prep time by **40%**.
- Mentored junior full stack engineers on backend API design and data contract principles, conducting code reviews on **Python** and **Node.js** services that consumed data platform outputs.
- Owned the technical design and execution of analytics infrastructure supporting payer and provider stakeholder reporting, balancing performance, security, and maintainability across 6 concurrent reporting domains.
- Led evaluation of transformation tooling alternatives, documenting trade-offs between **dbt**, **Coalesce**, and custom **Python** solutions and recommending **dbt** adoption for the healthcare engineering team based on deployment automation and testing capabilities.

Shopee

Jul 2018–Sep 2022

Full Stack Engineer

- Played a key role in building the e-commerce analytics platform supporting 295M+ users across Southeast Asia, designing **Snowflake** data warehouse schema for transaction, inventory, and seller performance data ingested via **Fivetran** from 20+ operational systems.
- Contributed to the medallion architecture implementation across Bronze (raw transaction feeds), Silver (normalized buyer/seller/product entities), and Gold (business KPIs) layers, establishing naming conventions and access patterns used by 40+ downstream analytics consumers.
- Implemented data transformation pipelines in **dbt** that processed 150M+ daily e-commerce transactions, seller shop closures, and product catalog changes, maintaining data freshness within 2-hour SLA for real-time inventory and demand analytics.
- Designed and built semantic models for marketplace KPIs including gross merchandising volume (GMV), take rate, seller cohort metrics, and customer lifetime value in **Snowflake**, enabling self-service analytics through **Tableau** dashboards.
- Built data quality and observability monitoring using **Python** and **Airflow** to track pipeline freshness, completeness, and anomalies across 50+ daily transformation jobs, catching schema drift and upstream data corruption issues before they propagated.
- Optimized **Snowflake** query performance for high-concurrency reporting workloads, tuning clustering keys for buyer transaction tables (300B+ rows) and implementing materialized views that reduced dashboard load times by **65%**.
- Contributed to data governance implementation, documenting PII classification for customer email/payment data, implementing access controls in **Snowflake** RBAC, and maintaining lineage documentation for compliance audits.
- Worked with Data Scientists to prepare feature datasets for demand forecasting and fraud detection models in **Python**, building **Spark**-based ETL jobs that prepared training and inference datasets for weekly model retraining cycles.

- Collaborated with Finance and GTM teams to expose marketplace financial data through **Tableau**, supporting revenue recognition reporting, seller payout audits, and acquisition cost analysis used by executive leadership.

Altitude Games

Nov 2014–Jul 2018

Full Stack Engineer

- Built backend data pipelines for F2P mobile games using **Python** and **PostgreSQL**, capturing player telemetry, in-game purchase events, and engagement metrics from 2M+ daily active users across casual and mid-core titles.
- Contributed to player analytics platform implementation, designing **PostgreSQL** schema to normalize player progression, session events, and monetization data that informed game balance and retention decisions.
- Implemented ETL jobs using **Python** and **Apache Spark** to aggregate daily player cohort metrics, churn predictions, and revenue trends, enabling analytics team to track game health across Kung Fu Clicker, Dream Defense, and Holy Ship! titles.
- Designed data integration from game servers to analytics warehouse, building **Kafka**-based event streaming architecture that guaranteed in-order delivery of player actions and purchase events with **99.5%** reliability.
- Collaborated with product and game design teams to instrument new telemetry events and expose player behavior data through dashboards, supporting A/B testing and feature decision-making for monetization experiments.
- Optimized **PostgreSQL** queries for daily reporting on player retention, lifetime value, and acquisition cost metrics, implementing indexes and materialized views that improved dashboard refresh times from 25 minutes to under **5 minutes**.

EDUCATION

Ateneo de Manila University

Aug 2010–Jul 2014

Bachelor of Science in Computer Science